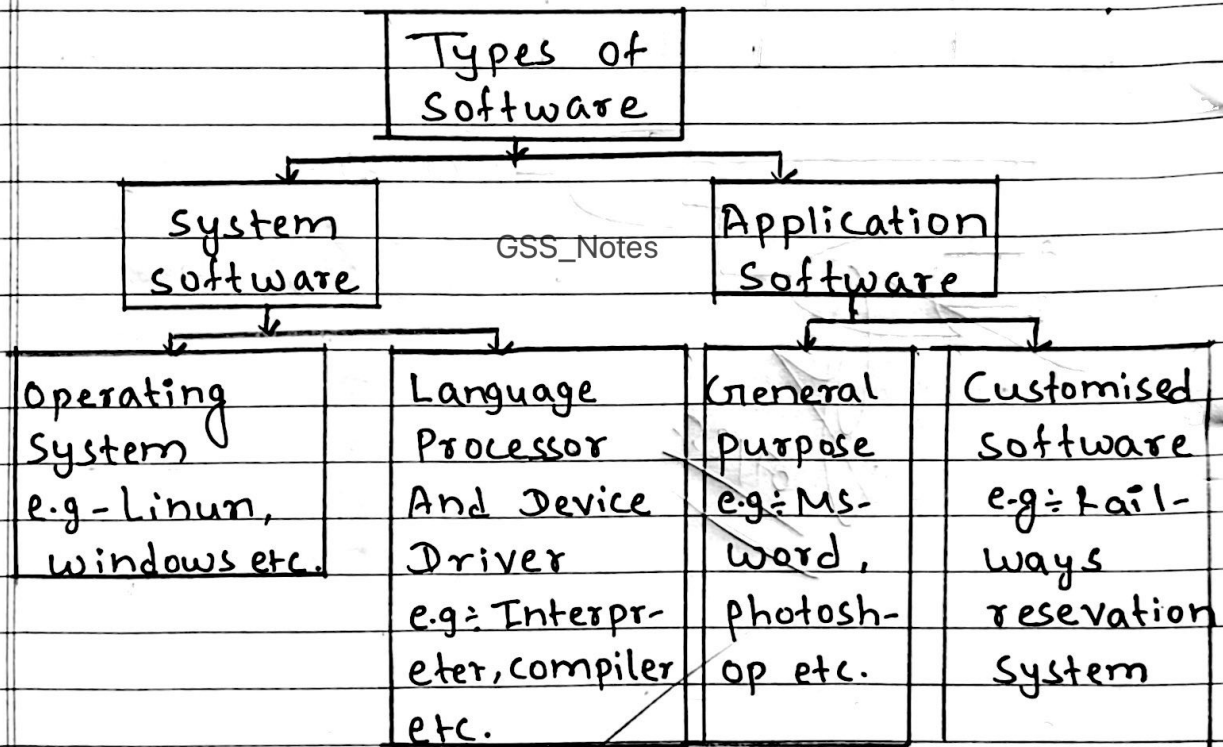


Chapter-3 Computer Software and Operating System

• Software

↳ The Software refers to the set of instructions fed in form of programs to govern the computer system and process the hardware components.



Software's can be divided into 3 major types:

1. System Software
2. Application Software
3. Utility Software

① System Software

- The System software is a collection of programs designed to operate, control, and extend the processing capabilities of the computer itself. System software is generally prepared by the computer manufacturers. System software serves as the interface between the hardware and the end user. Some examples of system software are operating system, compilers, interpreters, assemblers, etc.

② Operating System

- It is a collection of software that handles resources and provides general services for the other applications that run over them. As an OS essentially determines how a user interacts with the system, therefore many users prefer ~~to~~ to use one specific OS for their device. There are various types of operating system such as real-time, embedded, distributed, multiuser, single user, internet, mobile, and many more. It is important to consider the hardware specifications before choosing an operating system. Some examples of operating systems given below:

Android, iOS, Linux, Mac OS, MS windows, Ubuntu, Unix

⑥ Device Drivers

- It is a type of software that controls particular hardware which is attached to the system. Hardware devices that need a driver to connect to a system include displays, sound cards, printers, mice and harddisks. Further, there are two types of device drivers: Kernel Device Drivers and User Device Driver. Some examples of device drivers are:

- > Display Drivers
- > Motherboard Drivers
- > Printer Drivers
- > ROM Drivers
- > Sound Card Driver
- > USB Drivers
- > VGA Drivers

⑦ Language Translator

- A Translator or programming language processor is a generic term that can refer to anything that converts code from one computer language into another.

A. Compiler

→ A compiler translates the entire program written in a high-level language into machine code at once, before execution.

Examples of Compilers:

- C Compiler (GCC)
- C++ Compiler (Turbo C++)

B. Interpreter

→ An interpreter translates and executes one line of code written in high level language into machine understandable code at a time.

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If an error occurs, it stops immediately, allowing the programmer to fix the issue before continuing.

Examples: Python Interpreter.

C. Assembler

→ An assembler converts programs written in assembly language into machine code (object code).

→ Assembly language uses mnemonics (symbolic codes) such as MOV, ADD, SUB, etc, which correspond to binary instructions.

↳ Examples: MASM (Microsoft Macro Assembler)
TASM (Turbo Assembler)

• ~~Application Software~~

↳ ~~Application software is designed to perform~~
~~sp~~

① Utility Programs

↳ Utility Software helps maintain, manage, and protect the computer system. These tools perform specific maintenance tasks to ensure smooth operation.

↳ Common Functions

- Virus scanning and removal
- File compression and backup
- Disk cleanup and defragmentation
- Data recovery and system monitoring

↳ Examples:

- Antivirus: Avast, Norton, McAfee
- File management Tools: WinRAR, 7-Zip
- Backup Tools: Acronis, Google Drive Sync

② Firmware

↳ Firmware is a type of permanent software programmed directly into a hardware device (usually stored in ROM or fla-

sh memory).

↳ Purpose:

- Provides low-level control for hardware components.
- Starts hardware during boot-up (e.g. BIOS)
- Ensures the device functions as intended.

↳ Examples:

- BIOS / UEFI in Computers
- Firmware in routers, camera or smart TV

2. Application Software

↳ Application Software is designed to perform specific tasks for the user. Common types include general-purpose software (MS Word, Excel) and Customized software (school or hospital management systems).

It can be web-based, mobile, or desktop-based.

Application software increases productivity and makes work easier.

↳ Examples:

- Microsoft word, Excel (Productivity)
- Photoshop (Graphics)
- Chrome, Firefox (Web Browsers)
- Games, Media players

↳ Functions:

- Helps users perform tasks like writing, designing, calculating, browsing or communicating.

a. general - Purpose Application Software

↳ These are common programs used by most computer users to perform everyday tasks.

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Category	Description	Examples
• Word processing Software	Used to create, edit, and format text document	Ms word, Google Docs, WPS Writer
• Spreadsheet Software	Used for calculations, data analysis and charts.	Ms Excel, Google sheets, Libre office calc.
• Presentation Software	Used to create slideshows and visual presentations.	Ms powerpoint, Google slides, Canva

• Database Management System (DBMS)	Used to store, organize, and manage large amounts of data.	Ms Access, MySQL, Oracle DB
• Web browsers	Used to access and navigate the internet.	Google chrome, mozilla, Firefox, Safari
• Media players	Used to play audio and video files.	VLC Media Player, Windows Media, Player iTunes.

b. Specialized (or Custom) Application Software

→ These are designed for specific tasks or industries and may be custom-made for organization.

→ In Simple words:

- It is a tailor-made software designed for a specific company, task, or user requirement.

→ Example

- Healthcare Software, Banking software, educational Industry Software etc.

Operating System

• Operating System

→ An operating System (OS) is a system software that acts as an intermediary between computer hardware and users. It provides a user-friendly interface and manages computer hardware, software resources, and services for computer programs.

→ Usages of an Operating System:

> Resource Management

- Manages CPU, memory, and I/O devices.
e.g: Allocating CPU time for a running process.

> User interface

- Provides a graphical or command-line interface for interaction.
e.g: Windows desktop or Linux shell.

> Program Execution

- Loads and executes programs efficiently.
e.g: Running a web browser or a game.

> File management

- Organizes, stores, retrieves and manipulates data in storage
e.g: File Explorer in windows.

- > Security and Access Control
 - Protects data and resources from unauthorized access.
 - e.g: Login authentication in macOS.
- > Networking
 - Facilitates communication between devices over a network.
 - e.g - Sharing files using different protocols.

→ Functions of an Operating System

* Process management GSS_Notes

- > Definition - Manages processes by allocating resources, scheduling execution, and handling process termination.
- > Key features:
 - Supports multitasking by running multiple processes simultaneously.
 - Ensures fair CPU time allocation using scheduling algorithms.
 - Tracks process states (Ready, Running, waiting, etc)
 - Handles inter-process communication (IPC).

> Example

- Windows Task manager

* ~~Memory~~ Management

> Definition - Allocates and deallocates memory to processes, ensuring efficient memory utilization.

> Key features

- Provides Virtual memory to extend physical RAM.
- Prevents Memory leaks by reclaiming unused memory.
- Uses paging and segmentation for memory allocation.
- Protects memory spaces of different processes.

> Examples:

- Windows Memory Diagnostic Tool

* ~~File~~ System Management

> Definition - Manages file creation, deletion and access while organizing data on storage device.

> Key features

- Supports file hierarchies using directories and subdirectories.

- Ensures secure access with file permissions.
- Provides utilities for backup and recovery.
- Handles multiple file formats (e.g. NTFS, FAT32, ext4).
New Technology File System
File Allocating Table

> examples:

- Windows File Explorer

* Device Management

> Definition - Coordinates the functioning of hardware devices through device drivers.

> Key features

- Facilitates communication between OS and devices like printers, keyboards, and monitors.

- Ensures device availability using I/O scheduling.

- Monitors device performance and usage.

- Manages plug-and-play capabilities.

> Examples:

- Windows Device Manager

* Security and Protection

> Definition - Ensures the safety of data and resources by controlling access and preventing unauthorized activities.

> Key features:

- Provides authentication mechanisms like passwords and biometrics.

- Implements encryption for secure data transfer.

- Protects against malware and unauthorized access.

- Monitors and logs security events.

> examples

- Windows Defender

* User Interface Management

> Definition - Provides an interface for users to interact with the computer system, either through a GUI or CLI.

> Key Features:

- Enables graphical interaction using desk-

ops and icons.

- Offers command-line tools for advanced users.
- Supports customization of themes and layouts.
- Facilitates multitasking with multiple windows.

> Examples:

- Windows start Menu and Desktop
- GNOME and KDE on Linux

GSS_Notes

✓ Types of operating System

✓ 1. Batch Operating System

> Characteristics

- Jobs are grouped into batches.
- No direct interaction with the user.
- Jobs are executed sequentially.
- Suitable for repetitive tasks like payroll or data processing.
- Requires a job queue.

> Examples:

- IBM OS/360
- Batch jobs in UNIX

✓ ② Time-Sharing Operating System (Multitasking OS)

> characteristics:

- ↳ Multiple users share system resources simultaneously
- ↳ Time slots are allocated to each user (time-sharing).
- ↳ Improves CPU utilization
- ↳ Ensures responsiveness for users.
- ↳ Provides isolation among processes.

> Examples

- UNIX
- MULTICS

③ Real-Time Operating System (RTOS)

> characteristics:

- ↳ Processes data and provides output within a strict time constraint.
- ↳ Designed for embedded systems and critical applications.
- ↳ Hard RTOS: Strict timing constraints (e.g. - Pacemakers).
- ↳ Soft RTOS: Timing is critical but not life-threatening.

↳ ~~Priorit~~ Prioritizes tasks for timely execution.

> Examples:

- Vx Works
- FreeRTOS

④ Network Operating System

> characteristics:

↳ Manages network resources like servers and clients.

↳ Provides data security and resource sharing.

GSS_Notes

↳ Supports user access control and system monitoring.

↳ Centralized control for network administration.

↳ Operates on servers for multi-user collaboration.

> Examples

- Novell Netware
- Windows Server

⑤ Mobile Operating System

> Characteristics

- ↳ Designed for smartphones and tablets.
- ↳ Optimized for touch interfaces and mobile hardware.
- ↳ Supports app ecosystems and sensors.
- ↳ Focuses on low power consumption.
- ↳ Includes features like GPS and cellular connectivity.

> Examples:

- Android
- iOS

↙⑥ Multiprogramming Operating System

> Definition

- ↳ A type of OS that allows multiple programs to reside in memory simultaneously and executes them by switching between tasks to utilize CPU effectively.

> Major points:

- ↳ Maximizes CPU utilization
- The CPU always has a program to execute.

cure, reducing idle time.

↳ Job scheduling

- The OS uses job scheduling to decide which program to execute next.

↳ No Simultaneous Execution

- Only one process executes at a time, but others wait in memory.

↳ Efficient Resource Allocation

- Minimizes CPU and I/O device waiting times.

> Examples:

- IBM OS/360
- UNIX.

GSS_Notes

✓ ⑦ Multiprocessing Operating System

> Definition

- An OS that uses multiple processors to execute multiple processes simultaneously, increasing system performance and reliability.

> Major Points:

↳ Parallel processing

- Supports execution of multiple proce-

sses at the same time using different CPUs.

↳ Fault Tolerance

- If one processor fails, others continue to operate, providing high reliability.

↳ Improved Performance:

- Distributes tasks across processors, speeding up execution.

↳ Complex System Design

- Requires sophisticated coordination and communication among processors.

↳ Symmetric and Asymmetric Multiprocessing

- Symmetric: All processors share tasks equally (e.g. Windows).

- Asymmetric: A ~~major~~^{master} processor assigns tasks to slave processors (e.g. Older UNIX systems).

> Examples

- Linux (on multi-core processors)
- Windows Server
- IBM z10s

W(2) Multithreading Operating System

> Definition

- An OS that allows a single process to be divided into multiple threads, which can run concurrently to perform task more efficiently.

> Major Points

↳ Concurrent Task Execution

- Threads of a single process run concurrently, sharing resources like memory.

↳ Efficient Resource Use

- Threads share the same address space, reducing overhead compared to processes.

↳ Responsiveness

- Enables quicker application response by handling multiple threads.

↳ Simplified Communication

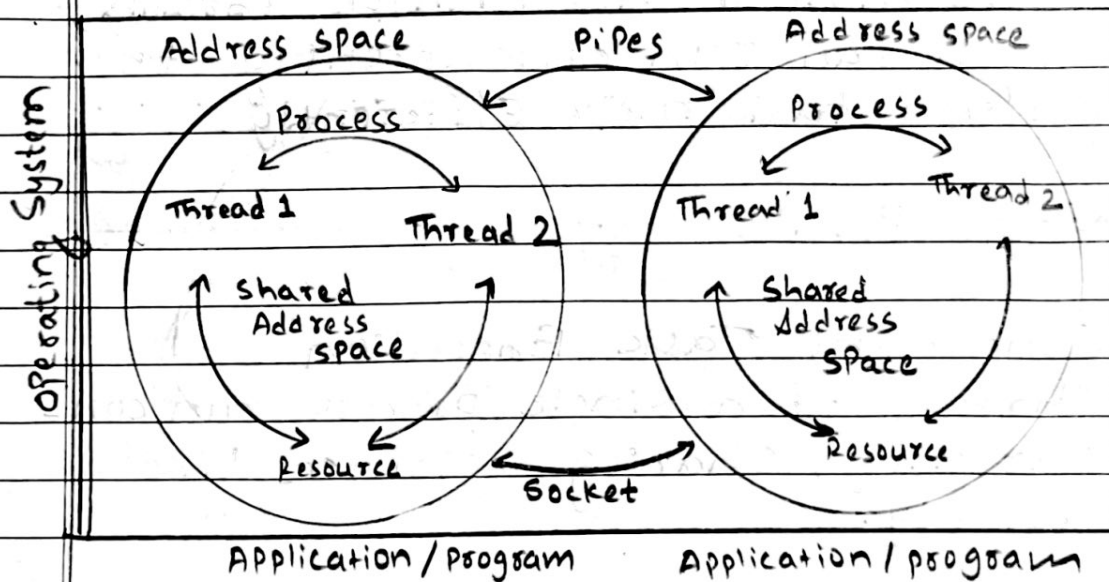
- Threads of the same process can easily communicate with each other.

↳ Suitable for Multi-Core Processors

- Utilizes multiple cores effectively to improve performance.

> Examples :

- Java Virtual Machine (JVM)
- Modern web browsers (e.g. google chrome)
- Apache HTTP Server



Assignment

- ① Write Seven differences between System Software and Application software.

System Software	Application Software
① A collection of programs designed to operate, control and extend the processing capabilities of computer itself is called system software.	A type of software designed to perform specific task for user is called Application Software.
② It is necessary for the computer to run.	It is not necessary for basic computer operation.
③ It runs in the background	It runs in the foreground when opened by the user.
④ It is installed when the operating system is installed	It is installed according to user needs.
⑤ It has direct access to hardware	It doesn't directly access hardware
⑥ It starts when the computer is powered on.	It starts when the user opens the program.

GSS_Notes

⑦ e.g ÷ Windows, Linum, e.g ÷ MS Word, Excel,
Mac Os Photoshop

② Write 5 differences between NTFS and FAT32

NTFS	FAT 32
① It is a modern file system	It is an older file system.
② It supports large file sizes (over 4GB)	It cannot store files larger than 4GB.
③ It provides file comp ression security and permissions.	It doesn't support file security.
④ It is more reliable and recovers from errors easily.	It is less reliable for error recovery.
⑤ It is mainly used in modern windows Systems.	It is widely used in USB drives and memory cards.

③ Write 7 differences between GUI and CUI

GUI	CUI
① GUI stands for Graphical User Interface.	CUI stands for Command User Interface.
② It uses graphics, icons and windows.	It uses text and commands.
③ It is easy to use.	It is difficult for beginners.
④ It uses mouse and touch input.	It mainly uses the keyboard.
⑤ It requires more memory and processing power.	It requires less memory and processing power.
⑥ It is commonly used in modern computers.	It is mostly used in older systems.
⑦ e.g.: Windows, macOS, Linux desktop.	e.g.: MS DOS, command Prompt.

- What is a Process in an Operating System?

→ In an Operating system, a process is an instance of a ~~process~~ program in execution. It is a basic unit of work that can be scheduled and executed by the operating system.

Any program that is executed is termed as a process. Processes change their state as they execute and can be either new, ready, running, waiting, or terminated. A process in OS is managed by the Process Control Block ~~(PCB)~~ (PCB).

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- Process Life Cycle

↳ The process life cycle is the sequence of states that a process goes through during its existence. The process life cycle can be divided into the following stages:

New - The process has been created but has not yet been scheduled for execution.

Ready - The process is ready to be executed.

Running - The process is currently being

executed.

Waiting - The process is waiting for an event to occur, such as the completion of an I/O operation. For example, the process may be waiting for keyboard input, disk access request, inter-process messages, a timer to go off, or a child process to finish.

Terminated - The process has finished executing

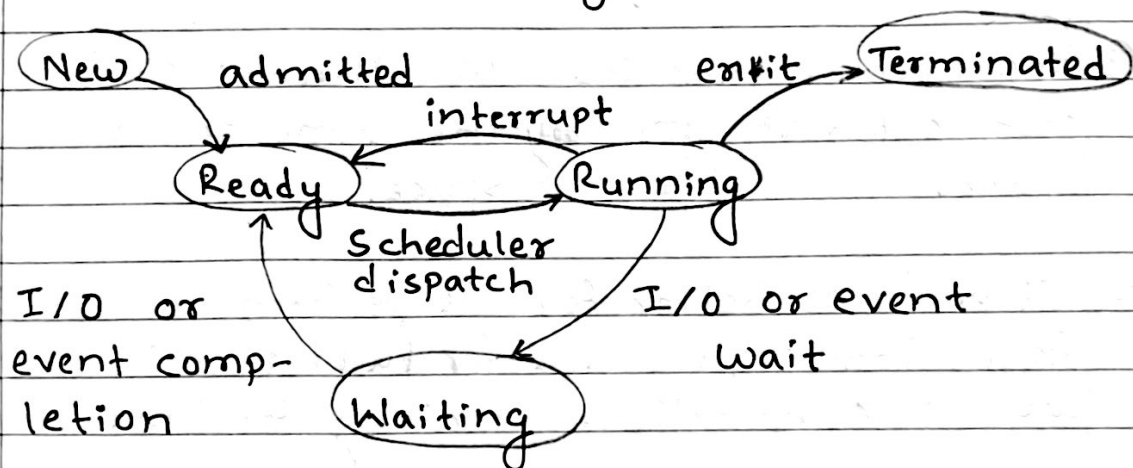


Figure - Diagram of process state

- Process example in (Google chrome)
 - ↳ Every tab = 1 process
 - ↳ Every extension = 1 process
 - ↳ chrome services = more processes
- Thread example in (Google chrome)
 - UI Thread, Network Thread

④ Write 7 differences between Multiprocessing and Multithreading Operating System (OS).

Multiprocessing OS	Multithreading OS
① Multiprocessing uses multiple CPU cores to run tasks at the same time.	Multithreading uses multiple threads that usually share the same CPU core.
② It runs several independent processes.	It runs several threads inside a single process.
③ Each process has its own separate memory space.	All threads share the same memory space.
④ It performs better for tasks that require heavy CPU computation.	It performs better for tasks that involve waiting for input or output.
⑤ Sharing data between processes is more difficult and needs special methods.	Sharing data between threads is easier because they already use the same memory.
⑥ If one process crashes, it usually does not affect other processes.	If one thread crashes, it can affect other threads in the same process.

Stop the other processes.

the entire program.

- ⑦ If it has higher overhead because every process needs its own resources. It has lower overhead because threads are lightweight.

- ⑧ Write seven differences between Open Source OS and proprietary OS.

Open Source OS

Proprietary OS

- ① An open source OS allows users to view, modify, and share its source code freely. A proprietary OS does not allow users to view or change the source code because it is owned by a company.

GSS_Notes

- ② Users can customize the system according to their needs. Users cannot customize the core system and must use it as provided by the company.

- ③ The software is usually free download and use. The software usually requires a paid license or purchase.

- | | |
|--|--|
| ④ Security issues are often fixed quickly because use many developers around the world can check the code. | Security fixes depend on the company, and users must wait for official updates. |
| ⑤ Users and communities provide support through forums and documentation. | Support is provided by the company through official help centers, updates, and customer service. |
| ⑥ Examples include Linux distribution such as Ubuntu or Debian. | Examples include Windows, macOS, and iOS. |
| ⑦ It is suitable for people who like control and flexibility over their system. | It is suitable for people who want a stable system that is managed and updated by a company. |

08/11/25